

Finding the first derivative of Sigmoid Function:

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$$Y = \frac{1}{1+e^{-z}} \quad (S^{-1}(y+1)) \quad \frac{b}{sb} \quad \frac{e^{-(y+1)}}{1+e^{-(y+1)}} \quad 1 - \frac{Y}{1-Y}$$

$$\frac{\partial Y}{\partial z} = \frac{\partial}{\partial z} \left(\frac{1}{1+e^{-z}} \right) \quad \frac{e^{-(y+1)}}{1+e^{-(y+1)}} \quad 1 - \frac{Y}{1-Y}$$

$$= - \frac{1}{(1+e^{-z})^2} \cdot (-e)^{-z} \quad \frac{e^{-y}}{S(y+1)}$$

$$= \frac{e^{-z}}{(1+e^{-z})^2} \quad \frac{1 - 1 + S^{-1}}{S(y+1)}$$

$$= \frac{1}{(1+e^{-z})} \cdot \frac{e^{-z}}{(1+e^{-z})} \quad \frac{1}{S^{-1} + 1}$$

$$= Y \cdot \frac{e^{-z} + 1 - 1}{(1+e^{-z})}$$

$$= Y \cdot \left(\frac{1+e^{-z}}{1+e^{-z}} - \frac{1}{1+e^{-z}} \right)$$

$$\therefore \frac{\partial Y}{\partial z} = Y \cdot (1 - Y)$$

$$\frac{\partial L}{\partial z} = \frac{\partial L}{\partial Y} \cdot \frac{\partial Y}{\partial z}$$

$$= (Y - T) \cdot Y(1 - Y)$$

$$\text{Loss, } L = \frac{1}{2} (T - Y)^2$$

Let, $u = T - Y$

$\therefore L = \frac{1}{2} u^2$

Using chain Rule,

$$\frac{\partial L}{\partial Y} = \frac{\partial L}{\partial u} \cdot \frac{\partial u}{\partial Y}$$

First, $\frac{\partial L}{\partial u} = \frac{\partial}{\partial u} \left(\frac{1}{2} u^2 \right) = u$

Second, $\frac{\partial u}{\partial Y} = \frac{\partial}{\partial Y} (T - Y) = -1$

$\therefore \frac{\partial L}{\partial Y} = \frac{\partial L}{\partial u} \cdot \frac{\partial u}{\partial Y}$

$= u \cdot (-1)$

$= -u$

$= -(T - Y)$

$= (Y - T)$

$\therefore \frac{\partial L}{\partial Y} = Y - T$

Ans